Lecture Scribe L9

L9_S1_A (4).pdf
PDF

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If any slide contains scribbles, handwritten notes, ink marks, highlights, circles, underlines, or annotations, ignore them completely. Use only the clean typed slide content as the source for the lecture scribe.

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All definitions, derivations, and examples correspond directly to the slides in the uploaded lecture file.

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Lecture Scribe

CSE400 – Fundamentals of Probability in Computing

Lecture 9: Uniform, Exponential, Laplace and Gamma Random Variables

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Outline

Types of Continuous Random Variables

- Uniform Random Variable: Example
- Exponential Random Variable: Example
- Laplace Random Variable: Example
- Gamma Random Variable
    - Graph and Special Cases
    - Example
    - Homework Problem

Problem Solving

Types of Continuous Random Variables

1. Uniform Random Variable

The probability density function (PDF) of a uniform random variable is

$$f_X(x) = \begin{cases} \dfrac{1}{b-a}, & a \le x < b \\ 0, & \text{elsewhere} \end{cases}$$

The cumulative distribution function (CDF) is

$$F_X(x) = \begin{cases} 0, & x < a \\ \dfrac{x-a}{b-a}, & a \le x < b \\ 1, & x \ge b \end{cases}$$

The PDF is constant over the interval $[a, b]$ and zero elsewhere.
The CDF increases linearly between a and b.

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Example 1 (Uniform Random Variable)

Problem

The phase of a sinusoid, θ, is uniformly distributed over $[0, 2\pi]$ with PDF

$$f_\theta(\theta) = \begin{cases} \dfrac{1}{2\pi}, & 0 \le \theta < 2\pi \\ 0, & \text{otherwise} \end{cases}$$

Find:

(a) $Pr(\theta > \frac{3\pi}{4})$

(b) $Pr(\theta < \pi \mid \theta > \frac{3\pi}{4})$

(c) $Pr(\cos(\theta) < \frac{1}{2})$

Solution

Given

$$f_\theta(\theta) = \begin{cases} \dfrac{1}{2\pi}, & 0 \le \theta < 2\pi \\ 0, & \text{otherwise} \end{cases}$$

For a uniform RV on $[0, 2\pi]$:

$$Pr(a < \theta < b) = \frac{b-a}{2\pi}$$

(a)

$$Pr\left(\theta > \frac{3\pi}{4}\right) = \frac{2\pi - \frac{3\pi}{4}}{2\pi} = \frac{5}{8}$$

(b)

Using conditional probability

$$Pr(A|B) = \frac{Pr(A \cap B)}{Pr(B)}$$

$$Pr\left(\frac{3\pi}{4} < \theta < \pi\right) = \frac{\pi - \frac{3\pi}{4}}{2\pi} = \frac{1}{8}$$

$$Pr(B) = Pr\left(\theta > \frac{3\pi}{4}\right) = \frac{5}{8}$$

$$Pr(\theta < \pi \mid \theta > \frac{3\pi}{4}) = \frac{1/8}{5/8} = \frac{1}{5}$$

(c)

$$\cos\theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}, \frac{5\pi}{3}$$

$$\cos\theta < \frac{1}{2} \quad \text{for} \quad \frac{\pi}{3} < \theta < \frac{5\pi}{3}$$

$$Pr\left(\cos\theta < \frac{1}{2}\right) = \frac{\frac{5\pi}{3} - \frac{\pi}{3}}{2\pi} = \frac{4\pi/3}{2\pi} = \frac{2}{3}$$

Uniform Random Variable — Applications

Examples include:

- Phase of a sinusoidal signal when all phase angles between 0 and 2π are equally likely.
- A random number generated between 0 and 1 for simulations.
- Arrival time of a user within a known time window when there is no time preference.

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2. Exponential Random Variable

For $b > 0$:

PDF

$$f_X(x) = \frac{1}{b} e^{-x/b} u(x)$$

CDF

$$F_X(x) = \left[1 - e^{-x/b}\right] u(x)$$

Example 2 (Exponential Random Variable)

Problem

Let X be an exponential random variable with

$$f_X(x) = e^{-x} u(x)$$

Find:

(a) $Pr(3X < 5)$

(b) $Pr(3X < y)$

(c) If $Y = 3X$, find $f_Y(y)$

Solution

Given

$$f_X(x) = e^{-x} u(x)$$

$$F_X(x) = \left[1 - e^{-x}\right] u(x)$$

(a)

$$Pr(3X < 5) = Pr\left(X < \frac{5}{3}\right)$$

$$= F_X\left(\frac{5}{3}\right) = 1 - e^{-5/3}$$

(b)

$$Pr(3X < y) = Pr\left(X < \frac{y}{3}\right)$$

$$= F_X\left(\frac{y}{3}\right) = \left[1 - e^{-y/3}\right] u(y)$$

(c)

Let

$$Y = 3X$$

From part (b):

$$F_Y(y) = \left[1 - e^{-y/3}\right] u(y)$$

Differentiate:

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

$$f_Y(y) = \frac{1}{3} e^{-y/3} u(y)$$

Thus Y is exponential with parameter

$$\lambda = \frac{1}{3}$$

Exponential Random Variable — Applications

- Time until a radioactive particle decays.
- Duration of an idle period of a wireless communication channel.
- Time until the next earthquake in a simplified seismic model.

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3. Laplace Random Variable

PDF

$$f_X(x) = \frac{1}{2b} \exp\left(-\frac{|x|}{b}\right)$$

CDF

$$F_X(x) = \begin{cases} \frac{1}{2} \exp\left(\frac{x}{b}\right), & x < 0 \\ 1 - \frac{1}{2} \exp\left(-\frac{x}{b}\right), & x \geq 0 \end{cases}$$

Application:

- Modeling the amplitude of a speech signal.

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Example 3 (Laplace Random Variable)

Problem

Let W be a Laplace random variable with

$$f_W(w) = ce^{-2|w|}$$

Find

(a) constant c

(b) $Pr(-1 < W < 2)$

(c) $Pr(W > 0 | -1 < W < 2)$

Solution

(a)

$$\int_{-\infty}^{\infty} ce^{-2|w|} dw = 1$$
$$= 2c \int_{0}^{\infty} e^{-2w} dw$$
$$= 2c \cdot \frac{1}{2}$$
$$= c$$

Hence

$$c = 1$$

(b)

$$Pr(-1 < W < 2) = \int_{-1}^{0} e^{2w} dw + \int_{0}^{2} e^{-2w} dw$$
$$= \frac{1}{2}(1 - e^{-2}) + \frac{1}{2}(1 - e^{-4})$$
$$= 1 - \frac{1}{2}(e^{-2} + e^{-4})$$

(c)

$$Pr(W > 0 | -1 < W < 2) = \frac{Pr(0 < W < 2)}{Pr(-1 < W < 2)}$$
$$Pr(0 < W < 2) = \int_{0}^{2} e^{-2w} dw = \frac{1}{2}(1 - e^{-4})$$
$$Pr(W > 0 | -1 < W < 2) = \frac{1 - e^{-4}}{2 - (e^{-2} + e^{-4})}$$

Laplace Random Variable — Applications

- Modeling impulsive noise in communication systems.

- Error distribution in L_1 regression.
- Noise used in differential privacy mechanisms.

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4. Gamma Random Variable

PDF

$$f_X(x) = \frac{(x/b)^{c-1}\exp(-x/b)}{b\Gamma(c)}u(x)$$

CDF

$$F_X(x) = \frac{\gamma(c, x/b)}{\Gamma(c)}u(x)$$

Definitions

Gamma Function

$$\Gamma(\alpha) = \int_0^\infty e^{-t}t^{\alpha-1}dt$$

Incomplete Gamma Function

$$\gamma(\alpha, \beta) = \int_0^\beta e^{-t}t^{\alpha-1}dt$$

Special Cases of Gamma Distribution

- If $b = 2$ and $c = 0.5$, the resulting RV is a **Chi-square random variable**.
- If $c = 1$, the resulting RV is an **Exponential random variable**.

where

$$k = c, \quad \theta = b$$

Example 4 (Gamma Random Variable)

Problem

Let

$$X \sim Gamma(\alpha, \lambda)$$

Find

- $E[X]$
- $Var(X)$

Solution

Given

$$f_X(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}, \quad x \geq 0$$

(a) Mean

$$E[X] = \int_0^\infty x f_X(x) dx$$

$$= \frac{1}{\lambda \Gamma(\alpha)} \int_0^\infty \lambda e^{-\lambda x} (\lambda x)^\alpha dx$$

$$= \frac{\Gamma(\alpha+1)}{\lambda \Gamma(\alpha)}$$

$$E[X] = \frac{\alpha}{\lambda}$$

(b) Variance

$$Var(X) = E[X^2] - (E[X])^2$$

First compute $E[X^2]$:

$$E[X^2] = \frac{\Gamma(\alpha+2)}{\lambda^2 \Gamma(\alpha)} = \frac{\alpha(\alpha+1)}{\lambda^2}$$

Thus

$$Var(X) = \frac{\alpha(\alpha+1)}{\lambda^2} - \left(\frac{\alpha}{\lambda}\right)^2$$

$$Var(X) = \frac{\alpha}{\lambda^2}$$

Example 5 (Gamma Random Variable)

Prove the following properties of the Gamma function:

(a)

$$\Gamma(n) = (n-1)!$$

for $n = 1, 2, 3, \ldots$

(b)

$$\Gamma(x+1) = x\Gamma(x)$$

(c)

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

Gamma Random Variable — Applications

- Total service time for multiple independent tasks where each task duration is exponential.
- Received signal energy computed from multiple signal samples.
- Waiting time until the k-th event in a Poisson process.

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Exercise: Problem Solving 1 (Movie Theater)

Suppose you plan to arrive at a movie theater with showtime T.

- Arriving s minutes early incurs cost $c \cdot s$.
- Arriving s minutes late incurs cost $k \cdot s$.

Travel time T_r has PDF $f(t)$ and CDF $F(t)$.

Arrival time

$$Arrival = d + T_r$$

Early arrival

$$cost = c(T - d - T_r)$$

Late arrival

$$cost = k(d + T_r - T)$$

Expected cost

$$E[C(d)] = \int_0^{T-d} c(T-d-t)f(t)dt + \int_{T-d}^{\infty} k(t+d-T)f(t)dt$$

Optimality condition

$$F(T-d^*) = \frac{k}{c+k}$$

Optimal departure time

$$d^* = T - F^{-1}\left(\frac{k}{c+k}\right)$$

Interpretation:

Depart such that

$$Pr(\text{arriving early}) = \frac{k}{c+k}$$

Exercise: Problem Solving 2 (CDF Analysis)

Given CDF

$$F_X(x) = \begin{cases} 0, & x < 0 \\ x^2, & 0 \le x \le 1 \\ 1, & x > 1 \end{cases}$$

PDF

$$f_X(x) = \frac{d}{dx} F_X(x) = 2x, \quad 0 \le x \le 1$$

Mean

$$\mu_X = \int_0^1 x(2x)dx = \frac{2}{3}$$

Variance

$$\sigma_X^2 = \int_0^1 x^2(2x)dx - \mu_X^2 = \frac{1}{2} - \frac{4}{9} = \frac{1}{18}$$

Skewness

$$c_s = \frac{E[(X-\mu)^3]}{\sigma^3} = -\frac{2\sqrt{2}}{5}$$

Kurtosis

$$c_k = \frac{E[(X-\mu)^4]}{\sigma^4} = \frac{12}{5}$$

Conclusion

- Distribution is **left-skewed**
- **Platykurtic** (lighter tails than Gaussian)

If you want, I can also convert this **entire lecture scribe into a ready-to-submit LaTeX document** (exactly formatted for Overleaf / assignment submission).

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Lecture Scribe

CSE400 – Fundamentals of Probability in Computing

Lecture 9: Uniform, Exponential, Laplace and Gamma Random Variables

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Outline

The lecture covers the following topics:

Types of Continuous Random Variables

- Uniform Random Variable: Example
- Exponential Random Variable: Example
- Laplace Random Variable: Example
- Gamma Random Variable
  - Graph and Special Cases
  - Example
  - Homework Problem

Problem Solving

L9_S1_A (4)

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1. Types of Continuous Random Variables

1.1 Uniform Random Variable

The uniform random variable is defined by the following PDF:

$$f_X(x) = \begin{cases} \dfrac{1}{b-a}, & a \le x < b, \\ 0, & \text{elsewhere} \end{cases}$$

Its CDF is

$$F_X(x) = \begin{cases} 0, & x < a, \\ \dfrac{x-a}{b-a}, & a \le x < b, \\ 1, & x \ge b. \end{cases}$$

The slides also present the graph of the PDF and the graph of the CDF for the uniform random variable.

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2. Example #1 (Uniform RV)

Problem

The phase of a sinusoid, Θ, is uniformly distributed over $[0, 2\pi)$ so that its PDF is

$$f_\Theta(\theta) = \begin{cases} \dfrac{1}{2\pi}, & 0 \le \theta < 2\pi, \\ 0, & \text{otherwise.} \end{cases}$$

Find:

- (a) $\Pr\left(\theta > \dfrac{3\pi}{4}\right)$
- (b) $\Pr\left(\theta < \pi \mid \theta > \dfrac{3\pi}{4}\right)$
- (c) $\Pr\left(\cos(\theta) < \dfrac{1}{2}\right)$ L9_S1_A (4)

Solution

Given

$$f_\Theta(\theta) = \begin{cases} \dfrac{1}{2\pi}, & 0 \le \theta < 2\pi, \\ 0, & \text{otherwise} \end{cases}$$

For a uniform RV on $[0, 2\pi)$,

$$\Pr(a < \Theta < b) = \frac{b-a}{2\pi}$$

(a) Compute $\Pr\left(\Theta > \dfrac{3\pi}{4}\right)$

$$\Pr\left(\Theta > \frac{3\pi}{4}\right) = \frac{2\pi - \frac{3\pi}{4}}{2\pi} = \frac{5}{8}$$

(b) Compute $\Pr\left(\Theta < \pi \mid \Theta > \frac{3\pi}{4}\right)$

Using

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}$$

the numerator is

$$\Pr\left(\frac{3\pi}{4} < \Theta < \pi\right) = \frac{\pi - \frac{3\pi}{4}}{2\pi} = \frac{1}{8}$$

and

$$\Pr(B) = \Pr\left(\Theta > \frac{3\pi}{4}\right) = \frac{5}{8}$$

Hence,

$$\Pr\left(\Theta < \pi \mid \Theta > \frac{3\pi}{4}\right) = \frac{1/8}{5/8} = \frac{1}{5}$$

(c) Compute $\Pr\left(\cos(\Theta) < \frac{1}{2}\right)$

$$\cos\Theta = \frac{1}{2} \Rightarrow \Theta = \frac{\pi}{3}, \ \frac{5\pi}{3}$$

Thus,

$$\cos\Theta < \frac{1}{2} \quad \text{for} \quad \frac{\pi}{3} < \Theta < \frac{5\pi}{3}$$

Therefore,

$$\Pr\left(\cos(\Theta) < \frac{1}{2}\right) = \frac{\frac{5\pi}{3} - \frac{\pi}{3}}{2\pi} = \frac{4\pi/3}{2\pi} = \frac{2}{3}$$

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3. Uniform Random Variable: Application Examples

The slides list the following application examples:

- The phase of a sinusoidal signal, when all phase angles between 0 and 2π are equally likely.
- A random number generated by a computer between 0 and 1 for simulations.
- The arrival time of a user within a known time window, assuming no time preference.

L9_S1_A (4)

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4. Exponential Random Variable

For any $b > 0$, the exponential random variable has PDF and CDF given by

$$f_X(x) = \frac{1}{b}\exp\left(-\frac{x}{b}\right)u(x)$$

and

$$F_X(x) = \left[1 - \exp\left(-\frac{x}{b}\right)\right]u(x)$$

The slides also present the graph of the PDF and the graph of the CDF for the exponential random variable, with the CDF graph labeled $(b = 2)$.

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5. Example #2 (Exponential RV)

Problem

Let X be an exponential random variable with PDF

$$f_X(x) = e^{-x}u(x)$$

Find:

- (a) $\Pr(3X < 5)$
- (b) Generalize your answer to part (a) to find $\Pr(3X < y)$ for some arbitrary constant y.
- (c) Note that if we define a new random variable according to $Y = 3X$ then your answer to part (b) is the CDF of Y, $F_Y(y)$. Given your answer to part (b), find $f_Y(y)$.

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Solution

Given

$$f_X(x) = e^{-x}u(x) \Rightarrow F_X(x) = \left[1 - e^{-x}\right]u(x)$$

(a) Compute $\Pr(3X < 5)$

$$\Pr(3X < 5) = \Pr\left(X < \frac{5}{3}\right) = F_X\left(\frac{5}{3}\right) = 1 - e^{-5/3}$$

(b) Compute $\Pr(3X < y)$

$$\Pr(3X < y) = \Pr\left(X < \frac{y}{3}\right)$$
$$= F_X\left(\frac{y}{3}\right) = \left[1 - e^{-y/3}\right]u(y)$$

(c) Let $Y = 3X$. Find $f_Y(y)$

From part (b), the CDF of Y is

$$F_Y(y) = \left[1 - e^{-y/3}\right] u(y)$$

Differentiating,

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{1}{3} e^{-y/3} u(y)$$

Hence, Y is an exponential random variable with parameter

$$\lambda = \frac{1}{3}$$

L9_S1_A (4)

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6. Exponential Random Variable: Application Examples

The slides list the following application examples:

- Time until a radioactive particle decays, assuming a constant decay rate.
- Duration of an idle period of a wireless communication channel before it becomes busy.
- Time until the next earthquake in a simplified seismic activity model with a constant occurrence rate.

L9_S1_A (4)

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7. Laplace Random Variable

The Laplace random variable is given by the following PDF:

$$f_X(x) = \frac{1}{2b} \exp\left(-\frac{|x|}{b}\right)$$

Its CDF is

$$F_X(x) = \begin{cases} \frac{1}{2} \exp\left(\frac{x}{b}\right), & x < 0, \\ 1 - \frac{1}{2} \exp\left(-\frac{x}{b}\right), & x \geq 0 \end{cases}$$

The slides state the following application:

The Laplace random variable has been used to model the amplitude of a speech (voice) signal.

The slides also present the graph of the PDF and the graph of the CDF for the Laplace random variable, with the CDF graph labeled $(b = 2)$.

L9_S1_A (4)

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8. Example #3 (Laplace RV)

Problem

Let W be a Laplace random variable with a PDF given by

$$f_W(w) = ce^{-2|w|}$$

Find:

- (a) the value of the constant c
- (b) $\Pr(-1 < W < 2)$
- (c) $\Pr(W > 0 \mid -1 < W < 2)$

L9_S1_A (4)

Solution

Given

$$f_W(w) = ce^{-2|w|}$$

(a) Find c

$$\int_{-\infty}^{\infty} ce^{-2|w|}\, dw = 2c \int_0^{\infty} e^{-2w}\, dw = 2c \cdot \frac{1}{2} = c$$

Therefore,

$$c = 1$$

(b) Find $\Pr(-1 < W < 2)$

$$\begin{aligned}\Pr(-1 < W < 2) &= \int_{-1}^{0} e^{2w}\, dw + \int_0^2 e^{-2w}\, dw \\ &= \frac{1}{2}(1 - e^{-2}) + \frac{1}{2}(1 - e^{-4}) \\ &= 1 - \frac{1}{2}(e^{-2} + e^{-4})\end{aligned}$$

(c) Find $\Pr(W > 0 \mid -1 < W < 2)$

$$\Pr(W > 0 \mid -1 < W < 2) = \frac{\Pr(0 < W < 2)}{\Pr(-1 < W < 2)}$$

$$\Pr(0 < W < 2) = \int_0^2 e^{-2w}\, dw = \frac{1}{2}(1 - e^{-4})$$

Also,

$$\Pr(-1 < W < 2) = 1 - \frac{1}{2}(e^{-2} + e^{-4})$$

Hence,

$$\Pr(W > 0 \mid -1 < W < 2) = \frac{1 - e^{-4}}{2 - (e^{-2} + e^{-4})}$$

L9_S1_A (4)

9. Laplace Random Variable: Application Examples

The slides list the following application examples:

- The modeling of impulsive noise in communication systems, where sudden large noise spikes occur more frequently than in light-tailed models.
- The error distribution in $L1$ (least absolute deviation) regression, where the Laplace distribution arises naturally when minimizing absolute error.
- The noise added in differential privacy mechanisms, where Laplace noise is used to protect sensitive database information while preserving statistical utility.

L9_S1_A (4)

10. Gamma Random Variable

The gamma random variable is defined by the following PDF and CDF:

$$f_X(x) = \frac{(x/b)^{c-1}\exp(-x/b)}{b\Gamma(c)}\,u(x)$$

$$F_X(x) = \frac{\gamma(c, x/b)}{\Gamma(c)}\,u(x)$$

Definitions

Gamma Function

$$\Gamma(\alpha) = \int_0^\infty e^{-t}t^{\alpha-1}\,dt$$

Incomplete Gamma Function

$$\gamma(\alpha, \beta) = \int_0^\beta e^{-t}t^{\alpha-1}\,dt$$

L9_S1_A (4)

11. Gamma Random Variable (Graphs and Special Cases)

The slides show graphs of the gamma PDF and the gamma CDF for multiple parameter choices.

The slides list the following resulting random variables:

- For $b = 2,\ c = 0.5$, the resulting RV is a Chi-square RV.
- For $c = 1$, the resulting RV is an Exponential RV.

The slide also states:

$$\text{where } k = c \text{ and } \theta = b$$

L9_S1_A (4)

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12. Example #4 (Gamma RV)

Problem

Let X be a gamma random variable with parameters α and λ. Calculate:

- (a) $E[X]$
- (b) $Var(X)$

L9_S1_A (4)

Solution

Given

$$X \sim Gamma(\alpha, \lambda)$$

with PDF

$$f_X(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}, \qquad x \geq 0$$

(a) Find $E[X]$

$$\begin{aligned} E[X] &= \int_0^\infty x f_X(x)\, dx = \frac{1}{\Gamma(\alpha)} \int_0^\infty \lambda x e^{-\lambda x} (\lambda x)^{\alpha-1}\, dx \\ &= \frac{1}{\lambda \Gamma(\alpha)} \int_0^\infty \lambda e^{-\lambda x} (\lambda x)^{\alpha}\, dx \\ &= \frac{\Gamma(\alpha+1)}{\lambda \Gamma(\alpha)} \qquad \text{(Textbook Eq. 6.1)} \\ E[X] &= \frac{\alpha}{\lambda} \end{aligned}$$

(b) Find $Var(X)$

$$Var(X) = E[X^2] - (E[X])^2$$

First compute $E[X^2]$:

$$\begin{aligned} E[X^2] &= \frac{1}{\Gamma(\alpha)} \int_0^\infty x^2 \lambda e^{-\lambda x} (\lambda x)^{\alpha-1}\, dx \\ &= \frac{1}{\lambda^2 \Gamma(\alpha)} \int_0^\infty \lambda e^{-\lambda x} (\lambda x)^{\alpha+1}\, dx \end{aligned}$$

$$= \frac{\Gamma(\alpha+2)}{\lambda^2\Gamma(\alpha)} = \frac{\alpha(\alpha+1)}{\lambda^2}$$

Thus,

$$Var(X) = \frac{\alpha(\alpha+1)}{\lambda^2} - \left(\frac{\alpha}{\lambda}\right)^2 = \frac{\alpha}{\lambda^2}$$

L9_S1_A (4)

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13. Example #5 (Gamma RV)

The slide states:

Prove the following properties of the Gamma function.

(a)

$$\Gamma(n) = (n-1)! \qquad \text{for } n = 1, 2, 3, \ldots$$

(b)

$$\Gamma(x+1) = x\Gamma(x)$$

(c)

$$\Gamma(1/2) = \sqrt{\pi}$$

The slide also states:

Post your answer on Campuswire!

L9_S1_A (4)

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14. Gamma Random Variable: Application Examples

The slides list the following application examples:

- The total service time for completing multiple independent tasks, where each task duration is exponentially distributed. The sum of k exponential stages follows a Gamma distribution.
- The received signal energy in detection systems, where the energy computed from multiple independent signal samples follows a Gamma (or Chi-square) distribution.
- The aggregate waiting time until the k-th event in a Poisson process, such as the time until the k-th packet arrival in a network.

L9_S1_A (4)

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15. Exercise: Problem Solving 1 (Movie Theater)

Optimization Problem

Suppose that you plan to arrive at a movie theater for a scheduled showtime at a fixed start time T.

- If you arrive s minutes early, you incur a waiting cost of $c \cdot s$.
- If you arrive s minutes late, you incur a missing scene cost of $k \cdot s$.
- Suppose also that the travel time from your home to the theater is a continuous random variable with probability density function $f(t)$.

Determine the departure time that minimizes your expected cost. L9_S1_A (4)

Solution

Let d denote the departure time from home. Let T_r be the random travel time with pdf $f(t)$ and CDF $F(t)$.

Arrival time

$$\text{Arrival} = d + T_r$$

Cost structure

Early arrival $(d + T_r < T)$: cost $= c(T - d - T_r)$

Late arrival $(d + T_r > T)$: cost $= k(d + T_r - T)$

Expected cost

$$E[C(d)] = \int_0^{T-d} c(T - d - t) f(t)\, dt + \int_{T-d}^{\infty} k(t + d - T) f(t)\, dt$$

Differentiate $E[C(d)]$ with respect to d and set to zero:

$$-cF(T - d) + k(1 - F(T - d)) = 0$$

Optimality condition

$$F(T - d^*) = \frac{k}{c + k}$$

Optimal departure time

$$d^* = T - F^{-1}\left(\frac{k}{c + k}\right)$$

The slide states:

Interpretation: Depart such that the probability of arriving early equals $\frac{k}{c + k}$. L9_S1_A (4)

---

16. Exercise: Problem Solving 2 (CDF Analysis)

The CDF of a RV is given by

$$F_X(x) = \begin{cases} 0, & x < 0 \\ x^2, & 0 \le x \le 1 \\ 1, & x > 1 \end{cases}$$

Find:

- Mean (μ_X)
- Variance (σ_X^2)
- Skewness (c_s)
- Kurtosis (c_k)

L9_S1_A (4)

Solution

Given CDF

$$F_X(x) = \begin{cases} 0, & x < 0 \\ x^2, & 0 \le x \le 1 \\ 1, & x > 1 \end{cases}$$

PDF

$$f_X(x) = \frac{d}{dx} F_X(x) = 2x, \qquad 0 \le x \le 1$$

Mean

$$\mu_X = \int_0^1 x(2x)\,dx = \frac{2}{3}$$

Variance

$$\sigma_X^2 = \int_0^1 x^2(2x)\,dx - \mu_X^2 = \frac{1}{2} - \frac{4}{9} = \frac{1}{18}$$

Skewness

$$c_s = \frac{E[(X-\mu)^3]}{\sigma^3} = -\frac{2\sqrt{2}}{5}$$

Kurtosis

$$c_k = \frac{E[(X-\mu)^4]}{\sigma^4} = \frac{12}{5}$$

Conclusion

- Distribution is left-skewed
- Platykurtic (lighter tails than Gaussian)

L9_S1_A (4)

17. Gaussian Modeling: From Noise to Math

Motivation

Why Probability? Because the World is Noisy

The Reality

- Sensors don't give "flat" lines (thermal noise, interference).
- Repeated measurements form a "cloud" of uncertainty.

The Simulation

- We model noise using the Gaussian RV: $X \sim \mathcal{N}(\mu, \sigma^2)$.
- Generative Formula: $X = \sigma Z + \mu$

The slide also states:

Coding Task 1: Transform standard randomness (Z) into a physical model (X).

The slide concludes:

Probability models the shape of uncertainty, not just exact values.

L9_S1_A (4)

---

18. Density Estimation: Learning from Data

Reality Check

We don't know μ or σ in the field.

The Problem

We only observe raw, noisy samples (a histogram).

The Solution

We use Sample Statistics to estimate the distribution.

The slide presents the following sequence:

$$\text{Raw Samples } \{x_i\} \rightarrow \text{Estimate } \hat{\mu}, \hat{\sigma} \rightarrow \text{Estimated PDF}$$

Key Concept

We estimate probability distributions, not deterministic functions. This "smooths" the noise into a predictive tool.

L9_S1_A (4)

---

19. Applications: Network Systems

In-Class Focus: Packet Delay Modeling

Case Study: 500 ICMP Pings (Latency Analysis)

Scenario

A server transmits packets. Delay varies due to queueing, congestion, and routing hops.

Engineering Goal

- Typical Delay
- Jitter (Variation)

The slide states:

Critical Discussion: Why is high jitter $(\hat{\sigma})$ more damaging to real-time video than a high constant delay $(\hat{\mu})$?

The slide also labels the figure as:

Histogram of observed latency.

L9_S1_A (4)

---

20. Applications: Broader Computing Contexts

Probability in Modern Engineering

Image Processing

Denoising: Every camera sensor has “grain.” We use Gaussian priors to distinguish between real light and random thermal noise.

Tech: Image Restorers

Cloud Systems

Tail Latency: In distributed systems, we model the “99th percentile” to ensure the slowest 1% of users still have a good experience.

Tech: SRE / DevOps

IoT & Robotics

Sensor Fusion: A GPS is never 100% accurate. Kalman filters use probability to “guess” the real location between noisy pings.

Tech: Autonomous Vehicles

The slide concludes:

From hardware sensors to software backends, uncertainty is the only constant.

L9_S1_A (4)

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Lecture Scribe

CSE400 – Fundamentals of Probability in Computing

Lecture 9: Uniform, Exponential, Laplace and Gamma Random Variables

L9_S1_A (4)

Outline

Types of Continuous Random Variables

- Uniform Random Variable: Example
- Exponential Random Variable: Example
- Laplace Random Variable: Example
- Gamma Random Variable
  - Graph and Special Cases
  - Example
  - Homework Problem

Problem Solving

L9_S1_A (4)

---

1. Types of Continuous Random Variables

1.1 Uniform Random Variable

The uniform random variable is defined by the following probability density function:

$$f_X(x) = \begin{cases} \dfrac{1}{b-a}, & a \le x < b, \\ 0, & \text{elsewhere} \end{cases}$$

The cumulative distribution function is

$$F_X(x) = \begin{cases} 0, & x < a, \\ \dfrac{x-a}{b-a}, & a \le x < b, \\ 1, & x \ge b \end{cases}$$

The slides also present the probability density function and cumulative distribution function graphs for a uniform random variable.

L9_S1_A (4)

---

2. Example #1 (Uniform RV)

Problem

The phase of a sinusoid, Θ, is uniformly distributed over $[0, 2\pi)$ so that its PDF is of the form

$$f_\Theta(\theta) = \begin{cases} \frac{1}{2\pi}, & 0 \le \theta < 2\pi, \\ 0, & \text{otherwise} \end{cases}$$

Find:

- (a) $\Pr\left(\theta > \frac{3\pi}{4}\right)$
- (b) $\Pr\left(\theta < \pi \mid \theta > \frac{3\pi}{4}\right)$
- (c) $\Pr\left(\cos(\theta) < \frac{1}{2}\right)$ L9_S1_A (4)

Solution

Given

$$f_\Theta(\theta) = \begin{cases} \frac{1}{2\pi}, & 0 \le \theta < 2\pi \\ 0, & \text{otherwise} \end{cases}$$

For a uniform RV on $[0, 2\pi)$,

$$\Pr(a < \Theta < b) = \frac{b-a}{2\pi}$$

(a) Find $\Pr\left(\Theta > \frac{3\pi}{4}\right)$

$$\Pr\left(\Theta > \frac{3\pi}{4}\right) = \frac{2\pi - \frac{3\pi}{4}}{2\pi} = \frac{5}{8}$$

(b) Find $\Pr\left(\Theta < \pi \mid \Theta > \frac{3\pi}{4}\right)$

Using

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}$$

$$\Pr\left(\frac{3\pi}{4} < \Theta < \pi\right) = \frac{\pi - \frac{3\pi}{4}}{2\pi} = \frac{1}{8}$$

$$\Pr(B) = \frac{5}{8}$$

Hence,

$$\Pr\left(\Theta < \pi \mid \Theta > \frac{3\pi}{4}\right) = \frac{1/8}{5/8} = \frac{1}{5}$$

(c) Find $\Pr\left(\cos(\Theta) < \frac{1}{2}\right)$

$$\cos\Theta = \frac{1}{2} \Rightarrow \Theta = \frac{\pi}{3}, \frac{5\pi}{3}$$

$$\cos\Theta < \frac{1}{2} \quad \text{for} \quad \frac{\pi}{3} < \Theta < \frac{5\pi}{3}$$

$$\Pr\left(\cos(\Theta) < \frac{1}{2}\right) = \frac{\frac{5\pi}{3} - \frac{\pi}{3}}{2\pi} = \frac{4\pi/3}{2\pi} = \frac{2}{3}$$

L9_S1_A (4)

---

3. Uniform Random Variable: Application Examples

The following application examples are listed:

- The phase of a sinusoidal signal, when all phase angles between 0 and 2π are equally likely.
- A random number generated by a computer between 0 and 1 for simulations.
- The arrival time of a user within a known time window, assuming no time preference.

L9_S1_A (4)

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4. Exponential Random Variable

The exponential random variable has a PDF and CDF given by, for any $b > 0$,

$$f_X(x) = \frac{1}{b}\exp\left(-\frac{x}{b}\right)u(x)$$

$$F_X(x) = \left[1 - \exp\left(-\frac{x}{b}\right)\right]u(x)$$

The slides also present the PDF graph and the CDF graph for the exponential random variable.

L9_S1_A (4)

---

5. Example #2 (Exponential RV)

Problem

Let X be an exponential random variable with PDF,

$$f_X(x) = e^{-x}u(x)$$

Find:

- (a) $\Pr(3X < 5)$

- (b) Generalize your answer to part (a) to find $\Pr(3X < y)$ for some arbitrary constant y.
- (c) Note that if we define a new random variable according to $Y = 3X$ then your answer to part (b) is the CDF of Y, $F_Y(y)$. Given your answer to part (b), find $f_Y(y)$.

L9_S1_A (4)

Solution

Given

$$f_X(x) = e^{-x}u(x) \Rightarrow F_X(x) = \left[1 - e^{-x}\right] u(x)$$

(a) Find $\Pr(3X < 5)$

$$\Pr(3X < 5) = \Pr\left(X < \frac{5}{3}\right) = F_X\left(\frac{5}{3}\right) = 1 - e^{-5/3}$$

(b) Find $\Pr(3X < y)$

$$\Pr(3X < y) = \Pr\left(X < \frac{y}{3}\right)$$
$$= F_X\left(\frac{y}{3}\right) = \left[1 - e^{-y/3}\right] u(y)$$

(c) Let $Y = 3X$. Find $f_Y(y)$

From part (b), the CDF of Y is

$$F_Y(y) = \left[1 - e^{-y/3}\right] u(y)$$

Differentiating,

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{1}{3} e^{-y/3} u(y)$$

Hence, Y is an exponential random variable with parameter

$$\lambda = \frac{1}{3}$$

L9_S1_A (4)

---

6. Exponential Random Variable: Application Examples

The following application examples are listed:

- Time until a radioactive particle decays, assuming a constant decay rate.
- Duration of an idle period of a wireless communication channel before it becomes busy.
- Time until the next earthquake in a simplified seismic activity model with a constant occurrence rate.

L9_S1_A (4)

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7. Laplace Random Variable

The Laplace random variable is defined by the following PDF:

$$f_X(x) = \frac{1}{2b} \exp\left(-\frac{|x|}{b}\right)$$

Its CDF is

$$F_X(x) = \begin{cases} \frac{1}{2} \exp\left(\frac{x}{b}\right), & x < 0, \\ 1 - \frac{1}{2} \exp\left(-\frac{x}{b}\right), & x \geq 0 \end{cases}$$

Application:

The Laplace random variable has been used to model the amplitude of a speech (voice) signal.

L9_S1_A (4)

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8. Example #3 (Laplace RV)

Problem

Let W be a Laplace random variable with a PDF given by

$$f_W(w) = ce^{-2|w|}$$

Find:

- (a) Find the value of the constant c.
- (b) Find $\Pr(-1 < W < 2)$.
- (c) Find $\Pr(W > 0 \mid -1 < W < 2)$.

L9_S1_A (4)

Solution

Given

$$f_W(w) = ce^{-2|w|}$$

(a) Find c

$$\int_{-\infty}^{\infty} ce^{-2|w|}\, dw = 2c \int_0^{\infty} e^{-2w}\, dw = 2c \cdot \frac{1}{2} = c$$

$$\Rightarrow c = 1$$

(b) Find $\Pr(-1 < W < 2)$

$$\Pr(-1 < W < 2) = \int_{-1}^{0} e^{2w}\, dw + \int_0^2 e^{-2w}\, dw$$

$$= \frac{1}{2}(1 - e^{-2}) + \frac{1}{2}(1 - e^{-4})$$

$$= 1 - \frac{1}{2}(e^{-2} + e^{-4})$$

(c) Find $\Pr(W > 0 \mid -1 < W < 2)$

$$\Pr(W > 0 \mid -1 < W < 2) = \frac{\Pr(0 < W < 2)}{\Pr(-1 < W < 2)}$$

$$\Pr(0 < W < 2) = \int_0^2 e^{-2w}\, dw = \frac{1}{2}(1 - e^{-4})$$

$$\Pr(-1 < W < 2) = 1 - \frac{1}{2}(e^{-2} + e^{-4})$$

Hence,

$$\Pr(W > 0 \mid -1 < W < 2) = \frac{1 - e^{-4}}{2 - (e^{-2} + e^{-4})}$$

L9_S1_A (4)

---

9. Laplace Random Variable: Application Examples

The following application examples are listed:

- The modeling of impulsive noise in communication systems, where sudden large noise spikes occur more frequently than in light-tailed models.
- The error distribution in L1 (least absolute deviation) regression, where the Laplace distribution arises naturally when minimizing absolute error.
- The noise added in differential privacy mechanisms, where Laplace noise is used to protect sensitive database information while preserving statistical utility.

L9_S1_A (4)

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10. Gamma Random Variable

The gamma random variable is defined by the following PDF and CDF:

$$f_X(x) = \frac{(x/b)^{c-1} \exp(-x/b)}{b\Gamma(c)} u(x)$$

$$F_X(x) = \frac{\gamma(c, x/b)}{\Gamma(c)} u(x)$$

Definitions

Gamma Function:

$$\Gamma(\alpha) = \int_0^\infty e^{-t} t^{\alpha-1} dt$$

Incomplete Gamma Function:

$$\gamma(\alpha, \beta) = \int_0^\beta e^{-t} t^{\alpha-1} dt$$

L9_S1_A (4)

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11. Gamma Random Variable (Graphs and Special Cases)

The slides present graphs of the gamma PDF and CDF for multiple parameter settings.

The following resulting random variables are stated:

- For $b = 2, c = 0.5$, the resulting RV is a Chi-square RV.
- For $c = 1$, the resulting RV is an Exponential RV.

where

$$k = c \quad \text{and} \quad \theta = b$$

L9_S1_A (4)

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12. Example #4 (Gamma RV)

Problem

Let X be a gamma random variable with parameters α and λ. Calculate:

- (a) $E[X]$
- (b) $Var(X)$

L9_S1_A (4)

Solution

Given

$$X \sim Gamma(\alpha, \lambda)$$

with PDF

$$f_X(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}, \qquad x \geq 0$$

(a) Find $E[X]$

$$E[X] = \int_0^\infty x f_X(x)\, dx = \frac{1}{\Gamma(\alpha)} \int_0^\infty \lambda x e^{-\lambda x} (\lambda x)^{\alpha-1} dx$$

$$= \frac{1}{\lambda\Gamma(\alpha)} \int_0^\infty \lambda e^{-\lambda x} (\lambda x)^\alpha dx$$

$$= \frac{\Gamma(\alpha+1)}{\lambda\Gamma(\alpha)} \qquad \text{(Textbook Eq. 6.1)}$$

$$E[X] = \frac{\alpha}{\lambda}$$

(b) Find $Var(X)$

$$Var(X) = E[X^2] - (E[X])^2$$

First compute $E[X^2]$:

$$E[X^2] = \frac{1}{\Gamma(\alpha)} \int_0^\infty x^2 \lambda e^{-\lambda x} (\lambda x)^{\alpha-1} dx$$

$$= \frac{1}{\lambda^2\Gamma(\alpha)} \int_0^\infty \lambda e^{-\lambda x} (\lambda x)^{\alpha+1} dx$$

$$= \frac{\Gamma(\alpha+2)}{\lambda^2\Gamma(\alpha)} = \frac{\alpha(\alpha+1)}{\lambda^2}$$

Thus,

$$Var(X) = \frac{\alpha(\alpha+1)}{\lambda^2} - \left(\frac{\alpha}{\lambda}\right)^2 = \frac{\alpha}{\lambda^2}$$

L9_S1_A (4)

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13. Example #5 (Gamma RV)

Prove the following properties of the Gamma function.

(a)

$$\Gamma(n) = (n-1)! \qquad \text{for } n = 1, 2, 3, \ldots$$

(b)

$$\Gamma(x+1) = x\Gamma(x)$$

(c)

$$\Gamma(1/2) = \sqrt{\pi}$$

Post your answer on Campuswire!

L9_S1_A (4)

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14. Gamma Random Variable: Application Examples

The following application examples are listed:

- The total service time for completing multiple independent tasks, where each task duration is exponentially distributed. The sum of k exponential stages follows a Gamma distribution.
- The received signal energy in detection systems, where the energy computed from multiple independent signal samples follows a Gamma (or Chi-square) distribution.
- The aggregate waiting time until the k-th event in a Poisson process, such as the time until the k-th packet arrival in a network.

L9_S1_A (4)

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15. Exercise: Problem Solving 1 (Movie Theater)

Optimization Problem

Suppose that you plan to arrive at a movie theater for a scheduled showtime at a fixed start time T.

- If you arrive s minutes early, you incur a waiting cost of $c \cdot s$.
- If you arrive s minutes late, you incur a missing scene cost of $k \cdot s$.
- Suppose also that the travel time from your home to the theater is a continuous random variable with probability density function $f(t)$.

Determine the departure time that minimizes your expected cost.

L9_S1_A (4)

Solution

Let d denote the departure time from home. Let T_r be the random travel time with pdf $f(t)$ and CDF $F(t)$.

Arrival time

$$\text{Arrival} = d + T_r$$

Cost structure

Early arrival $(d + T_r < T)$: cost $= c(T - d - T_r)$

Late arrival $(d + T_r > T)$: cost $= k(d + T_r - T)$

Expected cost

$$E[C(d)] = \int_0^{T-d} c(T - d - t) f(t)\, dt + \int_{T-d}^{\infty} k(t + d - T) f(t)\, dt$$

Differentiate $E[C(d)]$ with respect to d and set to zero:

$$-cF(T - d) + k(1 - F(T - d)) = 0$$

Optimality condition

$$F(T - d^*) = \frac{k}{c + k}$$

Optimal departure time

$$d^* = T - F^{-1}\left(\frac{k}{c + k}\right)$$

Interpretation:

Depart such that the probability of arriving early equals $\frac{k}{c + k}$. L9_S1_A (4)

---

16. Exercise: Problem Solving 2 (CDF Analysis)

The CDF of a RV is given by:

$$F_X(x) = \begin{cases} 0, & x < 0 \\ x^2, & 0 \le x \le 1 \\ 1, & x > 1 \end{cases}$$

Find:

- Mean (μ_X)
- Variance (σ_X^2)
- Skewness (c_s)
- Kurtosis (c_k) L9_S1_A (4)

Solution

Given CDF

$$F_X(x) = \begin{cases} 0, & x < 0 \\ x^2, & 0 \le x \le 1 \\ 1, & x > 1 \end{cases}$$

PDF

$$f_X(x) = \frac{d}{dx} F_X(x) = 2x, \qquad 0 \le x \le 1$$

Mean

$$\mu_X = \int_0^1 x(2x)\, dx = \frac{2}{3}$$

Variance

$$\sigma_X^2 = \int_0^1 x^2(2x)\,dx - \mu_X^2 = \frac{1}{2} - \frac{4}{9} = \frac{1}{18}$$

Skewness

$$c_s = \frac{E[(X-\mu)^3]}{\sigma^3} = -\frac{2\sqrt{2}}{5}$$

Kurtosis

$$c_k = \frac{E[(X-\mu)^4]}{\sigma^4} = \frac{12}{5}$$

Conclusion

- Distribution is left-skewed
- Platykurtic (lighter tails than Gaussian)

L9_S1_A (4)

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17. Gaussian Modeling: From Noise to Math

Motivation: Why Probability? Because the World is Noisy

The Reality

- Sensors don't give "flat" lines (thermal noise, interference).
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- We model noise using the Gaussian RV: $X \sim \mathcal{N}(\mu, \sigma^2)$.
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Coding Task 1:

Transform standard randomness (Z) into a physical model (X).

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L9_S1_A (4)

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Reality Check

We don't know μ or σ in the field.

The Problem

We only observe raw, noisy samples (a histogram).

The Solution

We use Sample Statistics to estimate the distribution.

The slide presents the sequence

$$\text{Raw Samples } \{x_i\} \rightarrow \text{Estimate } \hat{\mu}, \hat{\sigma} \rightarrow \text{Estimated PDF}$$

Key Concept

We estimate probability distributions, not deterministic functions. This “smooths” the noise into a predictive tool.

L9_S1_A (4)

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19. Applications: Network Systems

In-Class Focus: Packet Delay Modeling

Case Study: 500 ICMP Pings (Latency Analysis)

Scenario

A server transmits packets. Delay varies due to queueing, congestion, and routing hops.

Engineering Goal

- Typical Delay
- Jitter (Variation)

Critical Discussion:

Why is high jitter $(\hat{\sigma})$ more damaging to real-time video than a high constant delay $(\hat{\mu})$?

The figure is labeled:

Histogram of observed latency.

L9_S1_A (4)

---

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Probability in Modern Engineering

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Denoising:

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Cloud Systems

Tail Latency:

In distributed systems, we model the “99th percentile” to ensure the slowest 1% of users still have a good experience.

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Sensor Fusion:

A GPS is never 100% accurate. Kalman filters use probability to “guess” the real location between noisy pings.

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From hardware sensors to software backends, uncertainty is the only constant.

L9_S1_A (4)
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Lecture Scribe

CSE400 – Fundamentals of Probability in Computing

Lecture 9: Uniform, Exponential, Laplace and Gamma Random Variables

L9_S1_A (4)

Outline

Types of Continuous Random Variables

- Uniform Random Variable: Example
- Exponential Random Variable: Example
- Laplace Random Variable: Example
- Gamma Random Variable
    - Graph and Special Cases
    - Example
    - Homework Problem

Problem Solving

L9_S1_A (4)

---

1. Types of Continuous Random Variables

1.1 Uniform Random Variable

The uniform random variable is defined by the following probability density function:

$$f_X(x) = \begin{cases} \dfrac{1}{b-a}, & a \le x < b, \\ 0, & \text{elsewhere} \end{cases}$$

The cumulative distribution function is

$$F_X(x) = \begin{cases} 0, & x < a, \\ \dfrac{x-a}{b-a}, & a \le x < b, \\ 1, & x \ge b \end{cases}$$

The slides also present the probability density function and cumulative distribution function graphs for a uniform random variable.
L9_S1_A (4)

---

2. Example #1 (Uniform RV)

Problem

The phase of a sinusoid, Θ, is uniformly distributed over $[0, 2\pi)$ so that its PDF is of the form

$$f_\Theta(\theta) = \begin{cases} \dfrac{1}{2\pi}, & 0 \le \theta < 2\pi, \\ 0, & \text{otherwise} \end{cases}$$

Find:

- (a) $\Pr\left(\theta > \dfrac{3\pi}{4}\right)$
- (b) $\Pr\left(\theta < \pi \mid \theta > \dfrac{3\pi}{4}\right)$
- (c) $\Pr\left(\cos(\theta) < \dfrac{1}{2}\right)$ L9_S1_A (4)

Solution

Given

$$f_\Theta(\theta) = \begin{cases} \dfrac{1}{2\pi}, & 0 \le \theta < 2\pi \\ 0, & \text{otherwise} \end{cases}$$

For a uniform RV on $[0, 2\pi)$,

$$\Pr(a < \Theta < b) = \frac{b-a}{2\pi}$$

(a) Find $\Pr\left(\Theta > \frac{3\pi}{4}\right)$

$$\Pr\left(\Theta > \frac{3\pi}{4}\right) = \frac{2\pi - \frac{3\pi}{4}}{2\pi} = \frac{5}{8}$$

(b) Find $\Pr\left(\Theta < \pi \mid \Theta > \frac{3\pi}{4}\right)$

Using

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}$$

$$\Pr\left(\frac{3\pi}{4} < \Theta < \pi\right) = \frac{\pi - \frac{3\pi}{4}}{2\pi} = \frac{1}{8}$$

$$\Pr(B) = \frac{5}{8}$$

Hence,

$$\Pr\left(\Theta < \pi \mid \Theta > \frac{3\pi}{4}\right) = \frac{1/8}{5/8} = \frac{1}{5}$$

(c) Find $\Pr\left(\cos(\Theta) < \frac{1}{2}\right)$

$$\cos\Theta = \frac{1}{2} \Rightarrow \Theta = \frac{\pi}{3}, \frac{5\pi}{3}$$

$$\cos\Theta < \frac{1}{2} \quad \text{for} \quad \frac{\pi}{3} < \Theta < \frac{5\pi}{3}$$

$$\Pr\left(\cos(\Theta) < \frac{1}{2}\right) = \frac{\frac{5\pi}{3} - \frac{\pi}{3}}{2\pi} = \frac{4\pi/3}{2\pi} = \frac{2}{3}$$

L9_S1_A (4)

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3. Uniform Random Variable: Application Examples

The following application examples are listed:

- The phase of a sinusoidal signal, when all phase angles between 0 and 2π are equally likely.
- A random number generated by a computer between 0 and 1 for simulations.
- The arrival time of a user within a known time window, assuming no time preference.

L9_S1_A (4)

---

4. Exponential Random Variable

The exponential random variable has a PDF and CDF given by, for any $b > 0$,

$$f_X(x) = \frac{1}{b} \exp\left(-\frac{x}{b}\right) u(x)$$

$$F_X(x) = \left[1 - \exp\left(-\frac{x}{b}\right)\right] u(x)$$

The slides also present the PDF graph and the CDF graph for the exponential random variable.

L9_S1_A (4)

---

5. Example #2 (Exponential RV)

Problem

Let X be an exponential random variable with PDF,

$$f_X(x) = e^{-x} u(x)$$

Find:

- (a) $\Pr(3X < 5)$
- (b) Generalize your answer to part (a) to find $\Pr(3X < y)$ for some arbitrary constant y.
- (c) Note that if we define a new random variable according to $Y = 3X$ then your answer to part (b) is the CDF of Y, $F_Y(y)$. Given your answer to part (b), find $f_Y(y)$.

L9_S1_A (4)

Solution

Given

$$f_X(x) = e^{-x} u(x) \Rightarrow F_X(x) = \left[1 - e^{-x}\right] u(x)$$

(a) Find $\Pr(3X < 5)$

$$\Pr(3X < 5) = \Pr\left(X < \frac{5}{3}\right) = F_X\left(\frac{5}{3}\right) = 1 - e^{-5/3}$$

(b) Find $\Pr(3X < y)$

$$\Pr(3X < y) = \Pr\left(X < \frac{y}{3}\right)$$

$$= F_X\left(\frac{y}{3}\right) = \left[1 - e^{-y/3}\right] u(y)$$

(c) Let $Y = 3X$. Find $f_Y(y)$

From part (b), the CDF of Y is

$$F_Y(y) = \left[1 - e^{-y/3}\right] u(y)$$

Differentiating,

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{1}{3} e^{-y/3} u(y)$$

Hence, Y is an exponential random variable with parameter

$$\lambda = \frac{1}{3}$$

L9_S1_A (4)

---

6. Exponential Random Variable: Application Examples

The following application examples are listed:

- Time until a radioactive particle decays, assuming a constant decay rate.
- Duration of an idle period of a wireless communication channel before it becomes busy.
- Time until the next earthquake in a simplified seismic activity model with a constant occurrence rate.

L9_S1_A (4)

---

7. Laplace Random Variable

The Laplace random variable is defined by the following PDF:

$$f_X(x) = \frac{1}{2b} \exp\left(-\frac{|x|}{b}\right)$$

Its CDF is

$$F_X(x) = \begin{cases} \frac{1}{2} \exp\left(\frac{x}{b}\right), & x < 0, \\ 1 - \frac{1}{2} \exp\left(-\frac{x}{b}\right), & x \geq 0 \end{cases}$$

Application:

The Laplace random variable has been used to model the amplitude of a speech (voice) signal.

L9_S1_A (4)

---

8. Example #3 (Laplace RV)

Problem

Let W be a Laplace random variable with a PDF given by

$$f_W(w) = ce^{-2|w|}$$

Find:

- (a) Find the value of the constant c.

- (b) Find $\Pr(-1 < W < 2)$.
- (c) Find $\Pr(W > 0 \mid -1 < W < 2)$.

L9_S1_A (4)

Solution

Given

$$f_W(w) = ce^{-2|w|}$$

(a) Find c

$$\int_{-\infty}^{\infty} ce^{-2|w|}\, dw = 2c \int_0^{\infty} e^{-2w}\, dw = 2c \cdot \frac{1}{2} = c$$

$$\Rightarrow c = 1$$

(b) Find $\Pr(-1 < W < 2)$

$$\Pr(-1 < W < 2) = \int_{-1}^{0} e^{2w}\, dw + \int_0^2 e^{-2w}\, dw$$

$$= \frac{1}{2}(1 - e^{-2}) + \frac{1}{2}(1 - e^{-4})$$

$$= 1 - \frac{1}{2}(e^{-2} + e^{-4})$$

(c) Find $\Pr(W > 0 \mid -1 < W < 2)$

$$\Pr(W > 0 \mid -1 < W < 2) = \frac{\Pr(0 < W < 2)}{\Pr(-1 < W < 2)}$$

$$\Pr(0 < W < 2) = \int_0^2 e^{-2w}\, dw = \frac{1}{2}(1 - e^{-4})$$

$$\Pr(-1 < W < 2) = 1 - \frac{1}{2}(e^{-2} + e^{-4})$$

Hence,

$$\Pr(W > 0 \mid -1 < W < 2) = \frac{1 - e^{-4}}{2 - (e^{-2} + e^{-4})}$$

L9_S1_A (4)

---

9. Laplace Random Variable: Application Examples

The following application examples are listed:

- The modeling of impulsive noise in communication systems, where sudden large noise spikes occur more frequently than in light-tailed models.

- The error distribution in L1 (least absolute deviation) regression, where the Laplace distribution arises naturally when minimizing absolute error.
- The noise added in differential privacy mechanisms, where Laplace noise is used to protect sensitive database information while preserving statistical utility.

L9_S1_A (4)

---

10. Gamma Random Variable

The gamma random variable is defined by the following PDF and CDF:

$$f_X(x) = \frac{(x/b)^{c-1}\exp(-x/b)}{b\Gamma(c)}u(x)$$

$$F_X(x) = \frac{\gamma(c, x/b)}{\Gamma(c)}u(x)$$

Definitions

Gamma Function:

$$\Gamma(\alpha) = \int_0^\infty e^{-t}t^{\alpha-1}dt$$

Incomplete Gamma Function:

$$\gamma(\alpha, \beta) = \int_0^\beta e^{-t}t^{\alpha-1}dt$$

L9_S1_A (4)

---

11. Gamma Random Variable (Graphs and Special Cases)

The slides present graphs of the gamma PDF and CDF for multiple parameter settings.

The following resulting random variables are stated:

- For $b = 2, c = 0.5$, the resulting RV is a Chi-square RV.
- For $c = 1$, the resulting RV is an Exponential RV.

where

$$k = c \quad \text{and} \quad \theta = b$$

L9_S1_A (4)

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12. Example #4 (Gamma RV)

Problem

Let X be a gamma random variable with parameters α and λ. Calculate:

- (a) $E[X]$
- (b) $Var(X)$

L9_S1_A (4)

Solution

Given

$$X \sim Gamma(\alpha, \lambda)$$

with PDF

$$f_X(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}, \qquad x \geq 0$$

(a) Find $E[X]$

$$E[X] = \int_0^\infty x f_X(x)\, dx = \frac{1}{\Gamma(\alpha)} \int_0^\infty \lambda x e^{-\lambda x} (\lambda x)^{\alpha-1} dx$$

$$= \frac{1}{\lambda \Gamma(\alpha)} \int_0^\infty \lambda e^{-\lambda x} (\lambda x)^\alpha dx$$

$$= \frac{\Gamma(\alpha+1)}{\lambda \Gamma(\alpha)} \qquad \text{(Textbook Eq. 6.1)}$$

$$E[X] = \frac{\alpha}{\lambda}$$

(b) Find $Var(X)$

$$Var(X) = E[X^2] - (E[X])^2$$

First compute $E[X^2]$:

$$E[X^2] = \frac{1}{\Gamma(\alpha)} \int_0^\infty x^2 \lambda e^{-\lambda x} (\lambda x)^{\alpha-1} dx$$

$$= \frac{1}{\lambda^2 \Gamma(\alpha)} \int_0^\infty \lambda e^{-\lambda x} (\lambda x)^{\alpha+1} dx$$

$$= \frac{\Gamma(\alpha+2)}{\lambda^2 \Gamma(\alpha)} = \frac{\alpha(\alpha+1)}{\lambda^2}$$

Thus,

$$Var(X) = \frac{\alpha(\alpha+1)}{\lambda^2} - \left(\frac{\alpha}{\lambda}\right)^2 = \frac{\alpha}{\lambda^2}$$

L9_S1_A (4)

13. Example #5 (Gamma RV)

Prove the following properties of the Gamma function.

(a)

$$\Gamma(n) = (n-1)! \qquad \text{for } n = 1, 2, 3, \ldots$$

(b)

$$\Gamma(x+1) = x\Gamma(x)$$

(c)

$$\Gamma(1/2) = \sqrt{\pi}$$

Post your answer on Campuswire!

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14. Gamma Random Variable: Application Examples

The following application examples are listed:

- The total service time for completing multiple independent tasks, where each task duration is exponentially distributed. The sum of k exponential stages follows a Gamma distribution.
- The received signal energy in detection systems, where the energy computed from multiple independent signal samples follows a Gamma (or Chi-square) distribution.
- The aggregate waiting time until the k-th event in a Poisson process, such as the time until the k-th packet arrival in a network.

L9_S1_A (4)

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15. Exercise: Problem Solving 1 (Movie Theater)

Optimization Problem

Suppose that you plan to arrive at a movie theater for a scheduled showtime at a fixed start time T.

- If you arrive s minutes early, you incur a waiting cost of $c \cdot s$.
- If you arrive s minutes late, you incur a missing scene cost of $k \cdot s$.
- Suppose also that the travel time from your home to the theater is a continuous random variable with probability density function $f(t)$.

Determine the departure time that minimizes your expected cost.

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Solution

Let d denote the departure time from home. Let T_r be the random travel time with pdf $f(t)$ and CDF $F(t)$.

Arrival time

$$\text{Arrival} = d + T_r$$

Cost structure

Early arrival $(d + T_r < T)$: cost $= c(T - d - T_r)$

Late arrival $(d + T_r > T)$: cost $= k(d + T_r - T)$

Expected cost

$$E[C(d)] = \int_0^{T-d} c(T - d - t) f(t)\, dt + \int_{T-d}^{\infty} k(t + d - T) f(t)\, dt$$

Differentiate $E[C(d)]$ with respect to d and set to zero:

$$-cF(T - d) + k(1 - F(T - d)) = 0$$

Optimality condition

$$F(T - d^*) = \frac{k}{c + k}$$

Optimal departure time

$$d^* = T - F^{-1}\left(\frac{k}{c + k}\right)$$

Interpretation:

Depart such that the probability of arriving early equals $\frac{k}{c + k}$. L9_S1_A (4)

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16. Exercise: Problem Solving 2 (CDF Analysis)

The CDF of a RV is given by:

$$F_X(x) = \begin{cases} 0, & x < 0 \\ x^2, & 0 \le x \le 1 \\ 1, & x > 1 \end{cases}$$

Find:

- Mean (μ_X)
- Variance (σ_X^2)
- Skewness (c_s)
- Kurtosis (c_k) L9_S1_A (4)

Solution

Given CDF

$$F_X(x) = \begin{cases} 0, & x < 0 \\ x^2, & 0 \le x \le 1 \\ 1, & x > 1 \end{cases}$$

PDF

$$f_X(x) = \frac{d}{dx} F_X(x) = 2x, \qquad 0 \le x \le 1$$

Mean

$$\mu_X = \int_0^1 x(2x)\,dx = \frac{2}{3}$$

Variance

$$\sigma_X^2 = \int_0^1 x^2(2x)\,dx - \mu_X^2 = \frac{1}{2} - \frac{4}{9} = \frac{1}{18}$$

Skewness

$$c_s = \frac{E[(X-\mu)^3]}{\sigma^3} = -\frac{2\sqrt{2}}{5}$$

Kurtosis

$$c_k = \frac{E[(X-\mu)^4]}{\sigma^4} = \frac{12}{5}$$

Conclusion

- Distribution is left-skewed
- Platykurtic (lighter tails than Gaussian)

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17. Gaussian Modeling: From Noise to Math

Motivation: Why Probability? Because the World is Noisy

The Reality

- Sensors don't give "flat" lines (thermal noise, interference).
- Repeated measurements form a "cloud" of uncertainty.

The Simulation

- We model noise using the Gaussian RV: $X \sim \mathcal{N}(\mu, \sigma^2)$.
- Generative Formula: $X = \sigma Z + \mu$

Coding Task 1:

Transform standard randomness (Z) into a physical model (X).

$$\text{Probability models the shape of uncertainty, not just exact values.}$$

L9_S1_A (4)

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18. Density Estimation: Learning from Data

Reality Check

We don't know μ or σ in the field.

The Problem

We only observe raw, noisy samples (a histogram).

The Solution

We use Sample Statistics to estimate the distribution.

The slide presents the sequence

$$\text{Raw Samples } \{x_i\} \rightarrow \text{Estimate } \hat{\mu}, \hat{\sigma} \rightarrow \text{Estimated PDF}$$

Key Concept

We estimate probability distributions, not deterministic functions. This "smooths" the noise into a predictive tool.

L9_S1_A (4)

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19. Applications: Network Systems

In-Class Focus: Packet Delay Modeling

Case Study: 500 ICMP Pings (Latency Analysis)

Scenario

A server transmits packets. Delay varies due to queueing, congestion, and routing hops.

Engineering Goal

- Typical Delay
- Jitter (Variation)

Critical Discussion:

Why is high jitter $(\hat{\sigma})$ more damaging to real-time video than a high constant delay $(\hat{\mu})$?

The figure is labeled:

Histogram of observed latency.

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20. Applications: Broader Computing Contexts

Probability in Modern Engineering

Image Processing

Denoising:

Every camera sensor has “grain.” We use Gaussian priors to distinguish between real light and random thermal noise.

Tech: Image Restorers

Cloud Systems

Tail Latency:

In distributed systems, we model the “99th percentile” to ensure the slowest 1% of users still have a good experience.

Tech: SRE / DevOps

IoT & Robotics

Sensor Fusion:

A GPS is never 100% accurate. Kalman filters use probability to “guess” the real location between noisy pings.

Tech: Autonomous Vehicles

$$\text{From hardware sensors to software backends, uncertainty is the only constant.}$$

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Sources

Use ONLY the attached lecture slides / lecture PDF as context. Do NOT use any textbook or external knowledge.

If any slide contains scribbles, handwritten notes, ink marks, highlights, circles, underlines, or annotations, ignore them completely. Use only the clean typed slide content as the source for the lecture scribe.

The scribe must be a faithful reconstruction of what was taught in class. Every definition, notation,

assumption, result, proof step, derivation step, and example step must be directly traceable to the provided slides.

While writing the scribe, explicitly reason step by step through:

Definitions and notation

Assumptions and conditions

Statements of results or theorems

Proofs or derivations (only where they appear in the slides)

Worked examples (only where they appear in the slides), including all intermediate steps

Make all logical dependencies between concepts explicit so that a student can reliably revise using only this document.

Follow the lecture's original structure and scope. Preserve ordering of topics as presented.

Maintain a formal academic tone suitable for exam revision.

STRICT CONSTRAINTS:

Do NOT introduce intuition, interpretations, simplifications, or additional explanations.

Do NOT add new examples, proofs, or derivations.

Do NOT infer missing steps unless they are explicitly shown in the slides.

Do NOT use background knowledge.

Do NOT invent content.

Do NOT extract, interpret, or incorporate anything from handwritten or annotated regions.

This output is a lecture scribe for exam preparation, not a tutorial, summary, or creative explanation.

Lecture Scribe

CSE400 – Fundamentals of Probability in Computing

Lecture 9: Uniform, Exponential, Laplace and Gamma Random Variables

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Outline

Types of Continuous Random Variables

- Uniform Random Variable: Example
- Exponential Random Variable: Example
- Laplace Random Variable: Example
- Gamma Random Variable
  - Graph and Special Cases
  - Example
  - Homework Problem

Problem Solving

L9_S1_A (4)

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1. Types of Continuous Random Variables

1.1 Uniform Random Variable

The uniform random variable is given by the following PDF:

$$f_X(x) = \begin{cases} \dfrac{1}{b-a}, & a \leq x < b, \\ 0, & \text{elsewhere} \end{cases}$$

The CDF is

$$F_X(x) = \begin{cases} 0, & x < a, \\ \dfrac{x-a}{b-a}, & a \leq x < b, \\ 1, & x \geq b \end{cases}$$

The slides also show the probability density function and cumulative distribution function of a uniform random variable.

L9_S1_A (4)

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2. Example #1 (Uniform RV)

Problem

The phase of a sinusoid, Θ, is uniformly distributed over $[0, 2\pi)$ so that its PDF is of the form

$$f_\Theta(\theta) = \begin{cases} \dfrac{1}{2\pi}, & 0 \leq \theta < 2\pi, \\ 0, & \text{otherwise} \end{cases}$$

Find:

- (a) $\Pr\left(\theta > \frac{3\pi}{4}\right)$
- (b) $\Pr\left(\theta < \pi \mid \theta > \frac{3\pi}{4}\right)$
- (c) $\Pr\left(\cos(\theta) < \frac{1}{2}\right)$ L9_S1_A (4)

Solution

Given

$$f_\Theta(\theta) = \begin{cases} \frac{1}{2\pi}, & 0 \le \theta < 2\pi, \\ 0, & \text{otherwise} \end{cases}$$

For a uniform RV on $[0, 2\pi)$,

$$\Pr(a < \Theta < b) = \frac{b-a}{2\pi}$$

(a)

$$\Pr\left(\Theta > \frac{3\pi}{4}\right) = \frac{2\pi - \frac{3\pi}{4}}{2\pi} = \frac{5}{8}$$

(b)

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}$$

$$\Pr\left(\frac{3\pi}{4} < \Theta < \pi\right) = \frac{\pi - \frac{3\pi}{4}}{2\pi} = \frac{1}{8}$$

$$\Pr(B) = \Pr\left(\Theta > \frac{3\pi}{4}\right) = \frac{5}{8}$$

$$\Pr\left(\Theta < \pi \mid \Theta > \frac{3\pi}{4}\right) = \frac{1/8}{5/8} = \frac{1}{5}$$

(c)

$$\cos\Theta = \frac{1}{2} \Rightarrow \Theta = \frac{\pi}{3}, \frac{5\pi}{3}$$

$$\cos\Theta < \frac{1}{2} \quad \text{for} \quad \frac{\pi}{3} < \Theta < \frac{5\pi}{3}$$

$$\Pr\left(\cos(\Theta) < \frac{1}{2}\right) = \frac{\frac{5\pi}{3} - \frac{\pi}{3}}{2\pi} = \frac{4\pi/3}{2\pi} = \frac{2}{3}$$

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3. Uniform Random Variable: Application Examples

- The phase of a sinusoidal signal, when all phase angles between 0 and 2π are equally likely.
- A random number generated by a computer between 0 and 1 for simulations.
- The arrival time of a user within a known time window, assuming no time preference.

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4. Exponential Random Variable

The exponential random variable has a PDF and CDF given by, for any $b > 0$,

$$f_X(x) = \frac{1}{b} \exp\left(-\frac{x}{b}\right) u(x)$$

$$F_X(x) = \left[1 - \exp\left(-\frac{x}{b}\right)\right] u(x)$$

The slides also show the PDF and CDF graphs for the exponential random variable.

L9_S1_A (4)

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5. Example #2 (Exponential RV)

Problem

Let X be an exponential random variable with PDF,

$$f_X(x) = e^{-x} u(x)$$

Find:

- (a) $\Pr(3X < 5)$
- (b) Generalize your answer to part (a) to find $\Pr(3X < y)$ for some arbitrary constant y.
- (c) Note that if we define a new random variable according to $Y = 3X$ then your answer to part (b) is the CDF of Y, $F_Y(y)$. Given your answer to part (b), find $f_Y(y)$.

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Solution

Given

$$f_X(x) = e^{-x} u(x) \Rightarrow F_X(x) = \left[1 - e^{-x}\right] u(x)$$

(a)

$$\Pr(3X < 5) = \Pr\left(X < \frac{5}{3}\right) = F_X\left(\frac{5}{3}\right) = 1 - e^{-5/3}$$

(b)

$$\Pr(3X < y) = \Pr\left(X < \frac{y}{3}\right)$$

$$= F_X\left(\frac{y}{3}\right) = \left[1 - e^{-y/3}\right] u(y)$$

(c)

Let $Y = 3X$. From part (b), the CDF of Y is

$$F_Y(y) = \left[1 - e^{-y/3}\right] u(y)$$

Differentiating,

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{1}{3} e^{-y/3} u(y)$$

Hence, Y is an exponential random variable with parameter

$$\lambda = \frac{1}{3}$$

L9_S1_A (4)

---

6. Exponential Random Variable: Application Examples

- Time until a radioactive particle decays, assuming a constant decay rate.
- Duration of an idle period of a wireless communication channel before it becomes busy.
- Time until the next earthquake in a simplified seismic activity model with a constant occurrence rate.

L9_S1_A (4)

---

7. Laplace Random Variable

The Laplace random variable is given by the following PDF:

$$f_X(x) = \frac{1}{2b} \exp\left(-\frac{|x|}{b}\right)$$

Its CDF is

$$F_X(x) = \begin{cases} \frac{1}{2} \exp\left(\frac{x}{b}\right), & x < 0, \\ 1 - \frac{1}{2} \exp\left(-\frac{x}{b}\right), & x \geq 0 \end{cases}$$

Applications: The Laplace random variable has been used to model the amplitude of a speech (voice) signal.

L9_S1_A (4)

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8. Example #3 (Laplace RV)

Problem

Let W be a Laplace random variable with a PDF given by

$$f_W(w) = ce^{-2|w|}$$

Find:

- (a) Find the value of the constant c.
- (b) Find $\Pr(-1 < W < 2)$.
- (c) Find $\Pr(W > 0 \mid -1 < W < 2)$.

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Solution

Given

$$f_W(w) = ce^{-2|w|}$$

(a) Find c

$$\int_{-\infty}^{\infty} ce^{-2|w|}\,dw = 2c\int_0^{\infty} e^{-2w}\,dw = 2c\cdot\frac{1}{2} = c$$

$$\Rightarrow c = 1$$

(b) Find $\Pr(-1 < W < 2)$

$$\Pr(-1 < W < 2) = \int_{-1}^{0} e^{2w}\,dw + \int_0^2 e^{-2w}\,dw$$

$$= \frac{1}{2}(1 - e^{-2}) + \frac{1}{2}(1 - e^{-4})$$

$$= 1 - \frac{1}{2}(e^{-2} + e^{-4})$$

(c) Find $\Pr(W > 0 \mid -1 < W < 2)$

$$\Pr(W > 0 \mid -1 < W < 2) = \frac{\Pr(0 < W < 2)}{\Pr(-1 < W < 2)}$$

$$\Pr(0 < W < 2) = \int_0^2 e^{-2w}\,dw = \frac{1}{2}(1 - e^{-4})$$

$$\Pr(-1 < W < 2) = 1 - \frac{1}{2}(e^{-2} + e^{-4})$$

Hence,

$$\Pr(W > 0 \mid -1 < W < 2) = \frac{1 - e^{-4}}{2 - (e^{-2} + e^{-4})}$$

L9_S1_A (4)

9. Laplace Random Variable: Application Examples

- The modeling of impulsive noise in communication systems, where sudden large noise spikes occur more frequently than in light-tailed models.
- The error distribution in L1 (least absolute deviation) regression, where the Laplace distribution arises naturally when minimizing absolute error.
- The noise added in differential privacy mechanisms, where Laplace noise is used to protect sensitive database information while preserving statistical utility.

L9_S1_A (4)

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10. Gamma Random Variable

The gamma random variable is defined by the following PDF and CDF:

$$f_X(x) = \frac{(x/b)^{c-1}\exp(-x/b)}{b\Gamma(c)}u(x)$$

$$F_X(x) = \frac{\gamma(c, x/b)}{\Gamma(c)}u(x)$$

Definitions

Gamma Function:

$$\Gamma(\alpha) = \int_0^\infty e^{-t}t^{\alpha-1}dt$$

Incomplete Gamma Function:

$$\gamma(\alpha, \beta) = \int_0^\beta e^{-t}t^{\alpha-1}dt$$

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11. Gamma Random Variable (Graphs and Special Cases)

The slides present graphs of the gamma PDF and CDF for multiple parameter settings.

The following resulting random variables are stated:

- For $b = 2,\ c = 0.5$, the resulting RV is a Chi-square RV.
- For $c = 1$, the resulting RV is an Exponential RV.

where

$$k = c \qquad \text{and} \qquad \theta = b$$

L9_S1_A (4)

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12. Example #4 (Gamma RV)

Problem

Let X be a gamma random variable with parameters α and λ. Calculate:

- (a) $E[X]$
- (b) $Var(X)$ L9_S1_A (4)

Solution

Given

$$X \sim Gamma(\alpha, \lambda)$$

with PDF

$$f_X(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}, \qquad x \geq 0$$

(a) Find $E[X]$

$$E[X] = \int_0^\infty x f_X(x)\, dx = \frac{1}{\Gamma(\alpha)} \int_0^\infty \lambda x e^{-\lambda x} (\lambda x)^{\alpha-1} dx$$

$$= \frac{1}{\lambda \Gamma(\alpha)} \int_0^\infty \lambda e^{-\lambda x} (\lambda x)^\alpha dx$$

$$= \frac{\Gamma(\alpha+1)}{\lambda \Gamma(\alpha)} \qquad \text{(Textbook Eq. 6.1)}$$

$$E[X] = \frac{\alpha}{\lambda}$$

(b) Find $Var(X)$

$$Var(X) = E[X^2] - (E[X])^2$$

First compute $E[X^2]$:

$$E[X^2] = \frac{1}{\Gamma(\alpha)} \int_0^\infty x^2 \lambda e^{-\lambda x} (\lambda x)^{\alpha-1} dx$$

$$= \frac{1}{\lambda^2 \Gamma(\alpha)} \int_0^\infty \lambda e^{-\lambda x} (\lambda x)^{\alpha+1} dx$$

$$= \frac{\Gamma(\alpha+2)}{\lambda^2 \Gamma(\alpha)} = \frac{\alpha(\alpha+1)}{\lambda^2}$$

Thus,

$$Var(X) = \frac{\alpha(\alpha+1)}{\lambda^2} - \left(\frac{\alpha}{\lambda}\right)^2 = \frac{\alpha}{\lambda^2}$$

L9_S1_A (4)

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13. Example #5 (Gamma RV)

Prove the following properties of the Gamma function.

(a)

$$\Gamma(n) = (n-1)! \qquad \text{for } n = 1, 2, 3, \ldots$$

(b)

$$\Gamma(x+1) = x\Gamma(x)$$

(c)

$$\Gamma(1/2) = \sqrt{\pi}$$

Post your answer on Campuswire!

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14. Gamma Random Variable: Application Examples

- The total service time for completing multiple independent tasks, where each task duration is exponentially distributed. The sum of k exponential stages follows a Gamma distribution.
- The received signal energy in detection systems, where the energy computed from multiple independent signal samples follows a Gamma (or Chi-square) distribution.
- The aggregate waiting time until the k-th event in a Poisson process, such as the time until the k-th packet arrival in a network.

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15. Exercise: Problem Solving 1 (Movie Theater)

Optimization Problem

Suppose that you plan to arrive at a movie theater for a scheduled showtime at a fixed start time T.

- If you arrive s minutes early, you incur a waiting cost of $c \cdot s$.
- If you arrive s minutes late, you incur a missing scene cost of $k \cdot s$.
- Suppose also that the travel time from your home to the theater is a continuous random variable with probability density function $f(t)$.

Determine the departure time that minimizes your expected cost.

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Solution

Let d denote the departure time from home. Let T_r be the random travel time with pdf $f(t)$ and CDF $F(t)$.

Arrival time

$$\text{Arrival} = d + T_r$$

Cost structure

Early arrival $(d + T_r < T)$: cost $= c(T - d - T_r)$

Late arrival $(d + T_r > T)$: cost $= k(d + T_r - T)$

Expected cost

$$E[C(d)] = \int_0^{T-d} c(T - d - t) f(t)\, dt + \int_{T-d}^{\infty} k(t + d - T) f(t)\, dt$$

Differentiate $E[C(d)]$ with respect to d and set to zero:

$$-cF(T - d) + k(1 - F(T - d)) = 0$$

Optimality condition

$$F(T - d^*) = \frac{k}{c + k}$$

Optimal departure time

$$d^* = T - F^{-1}\left(\frac{k}{c + k}\right)$$

Interpretation: Depart such that the probability of arriving early equals $\frac{k}{c + k}$. L9_S1_A (4)

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16. Exercise: Problem Solving 2 (CDF Analysis)

The CDF of a RV is given by

$$F_X(x) = \begin{cases} 0, & x < 0 \\ x^2, & 0 \le x \le 1 \\ 1, & x > 1 \end{cases}$$

Find:

- Mean (μ_X)
- Variance (σ_X^2)
- Skewness (c_s)
- Kurtosis (c_k)

Solution

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Given CDF

$$F_X(x) = \begin{cases} 0, & x < 0 \\ x^2, & 0 \le x \le 1 \\ 1, & x > 1 \end{cases}$$

PDF

$$f_X(x) = \frac{d}{dx} F_X(x) = 2x, \qquad 0 \le x \le 1$$

Mean

$$\mu_X = \int_0^1 x(2x)\,dx = \frac{2}{3}$$

Variance

$$\sigma_X^2 = \int_0^1 x^2(2x)\,dx - \mu_X^2 = \frac{1}{2} - \frac{4}{9} = \frac{1}{18}$$

Skewness

$$c_s = \frac{E[(X-\mu)^3]}{\sigma^3} = -\frac{2\sqrt{2}}{5}$$

Kurtosis

$$c_k = \frac{E[(X-\mu)^4]}{\sigma^4} = \frac{12}{5}$$

Conclusion

- Distribution is left-skewed
- Platykurtic (lighter tails than Gaussian)

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17. Gaussian Modeling: From Noise to Math

Motivation: Why Probability? Because the World is Noisy

The Reality

- Sensors don't give "flat" lines (thermal noise, interference).
- Repeated measurements form a "cloud" of uncertainty.

The Simulation

- We model noise using the Gaussian RV: $X \sim \mathcal{N}(\mu, \sigma^2)$.
- Generative Formula: $X = \sigma Z + \mu$

Coding Task 1: Transform standard randomness (Z) into a physical model (X).

$$\text{Probability models the shape of uncertainty, not just exact values.}$$

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18. Density Estimation: Learning from Data

Reality Check

We don't know μ or σ in the field.

The Problem

We only observe raw, noisy samples (a histogram).

The Solution

We use Sample Statistics to estimate the distribution.

The slide presents the sequence

$$\text{Raw Samples } \{x_i\} \rightarrow \text{Estimate } \hat{\mu}, \hat{\sigma} \rightarrow \text{Estimated PDF}$$

Key Concept

We estimate probability distributions, not deterministic functions. This "smooths" the noise into a predictive tool.

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19. Applications: Network Systems

In-Class Focus: Packet Delay Modeling

Case Study: 500 ICMP Pings (Latency Analysis)

Scenario

A server transmits packets. Delay varies due to queueing, congestion, and routing hops.

Engineering Goal

- Typical Delay
- Jitter (Variation)

Critical Discussion: Why is high jitter $(\hat{\sigma})$ more damaging to real-time video than a high constant delay $(\hat{\mu})$?

Figure: Histogram of observed latency.

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20. Applications: Broader Computing Contexts

Probability in Modern Engineering

Image Processing

Denoising: Every camera sensor has “grain.” We use Gaussian priors to distinguish between real light and random thermal noise.

Tech: Image Restorers

Cloud Systems

Tail Latency: In distributed systems, we model the “99th percentile” to ensure the slowest 1% of users still have a good experience.

Tech: SRE / DevOps

IoT & Robotics

Sensor Fusion: A GPS is never 100% accurate. Kalman filters use probability to “guess” the real location between noisy pings.

Tech: Autonomous Vehicles

$$\text{From hardware sensors to software backends, uncertainty is the only constant.}$$

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Sources